Short Course Template
8 wide by 9 high

L2 starts here

Multi AI Agent Systems with crewAI

Create agents to research and write an article



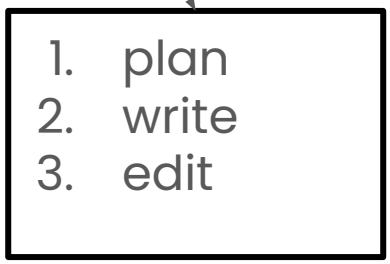


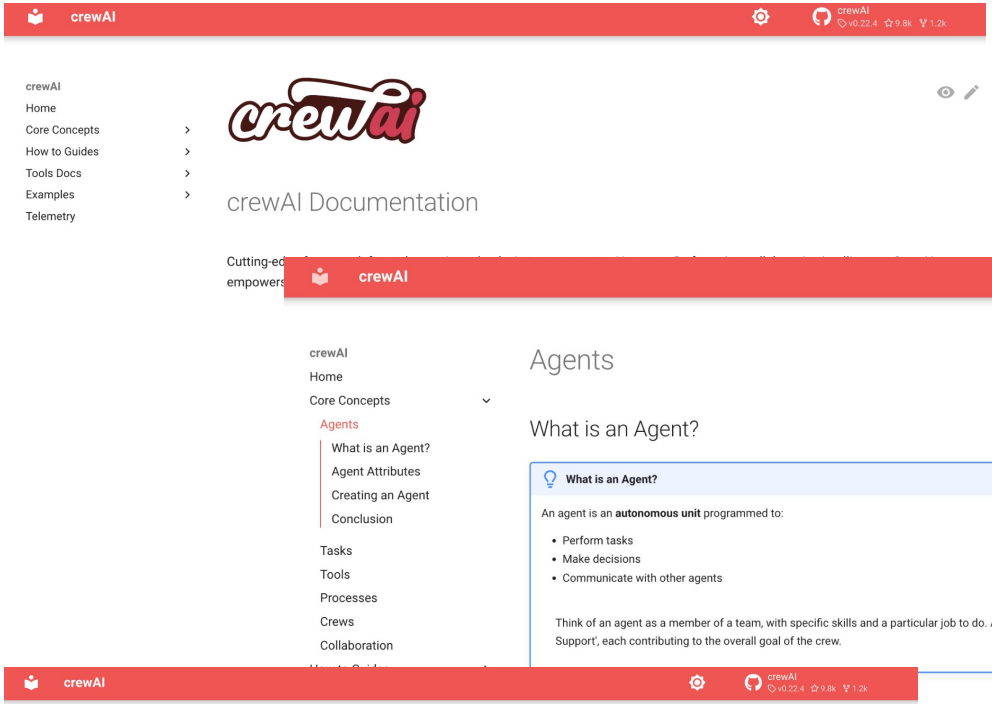
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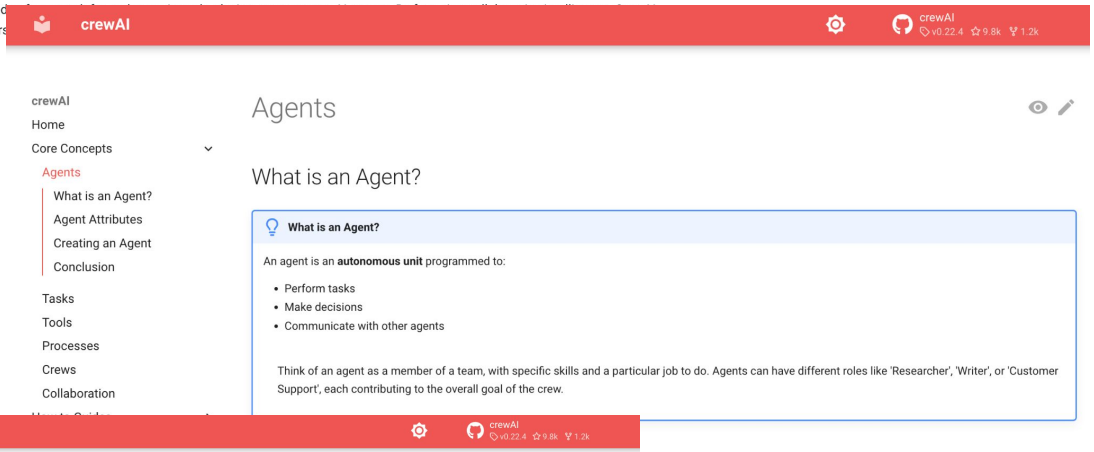
[Jupyter Notebook]

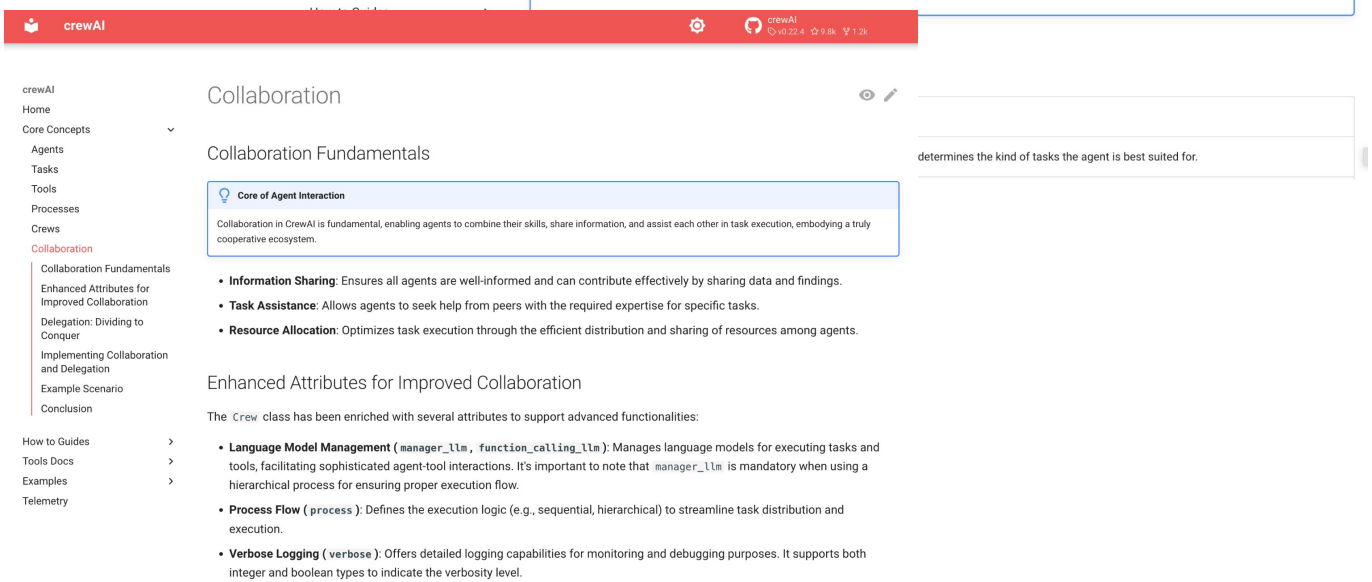
Tasks are Sequential by Default

```
crew = Crew(
    agents=[planner, writer, editor],
    tasks=[plan, write, edit],
    verbose=2
)
```

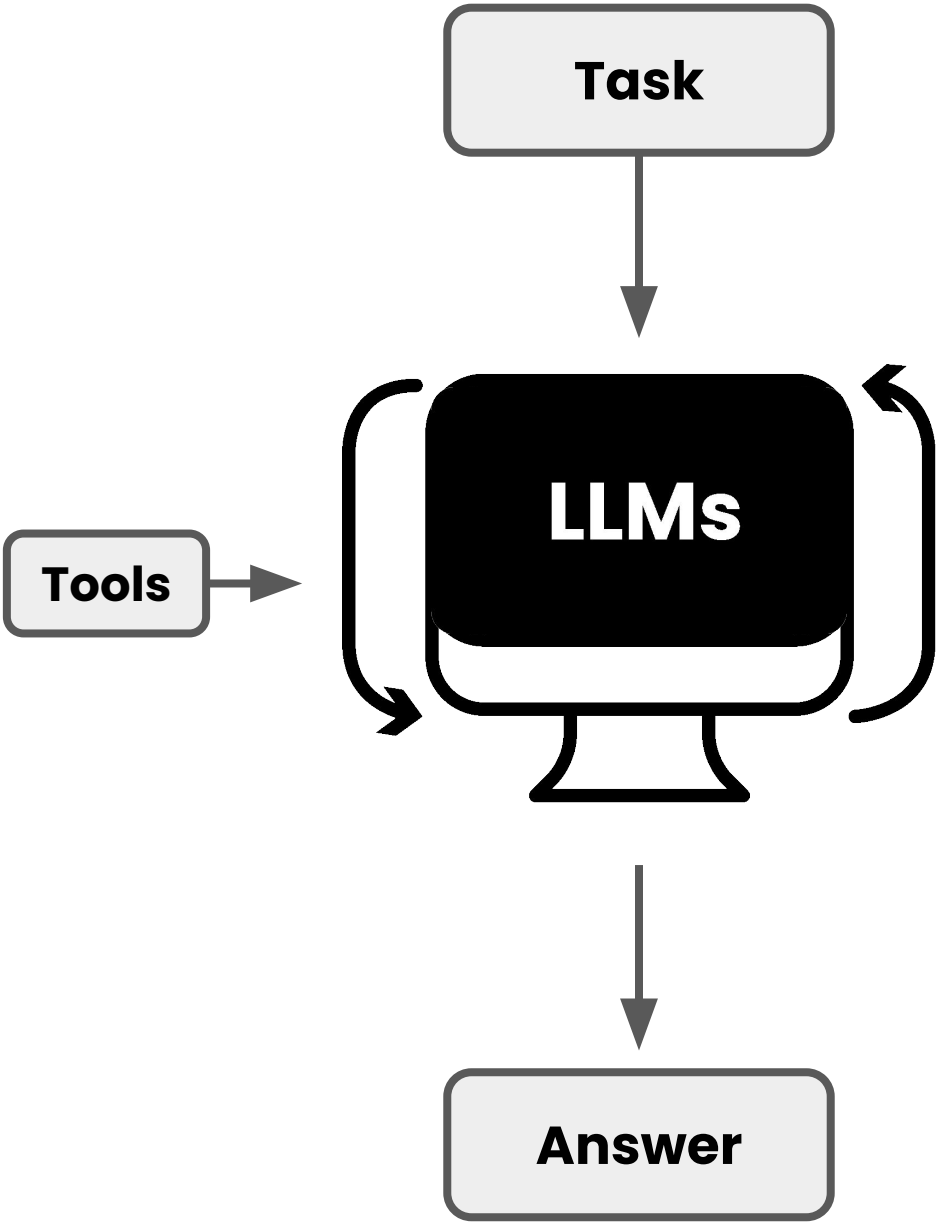


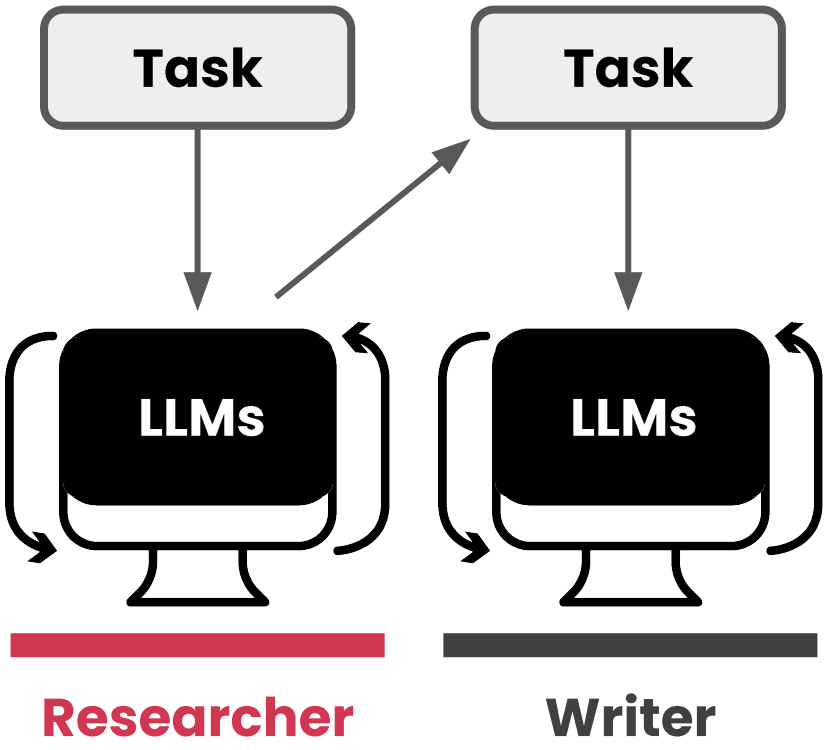

crewAI
crewAI
v0.22.4 9.8k 1.2k
crewAI
Home
Core Concepts
How to Guides
Tools Docs
Examples
Telemetry
crewAI Documentation
Cutting-ed
empowers

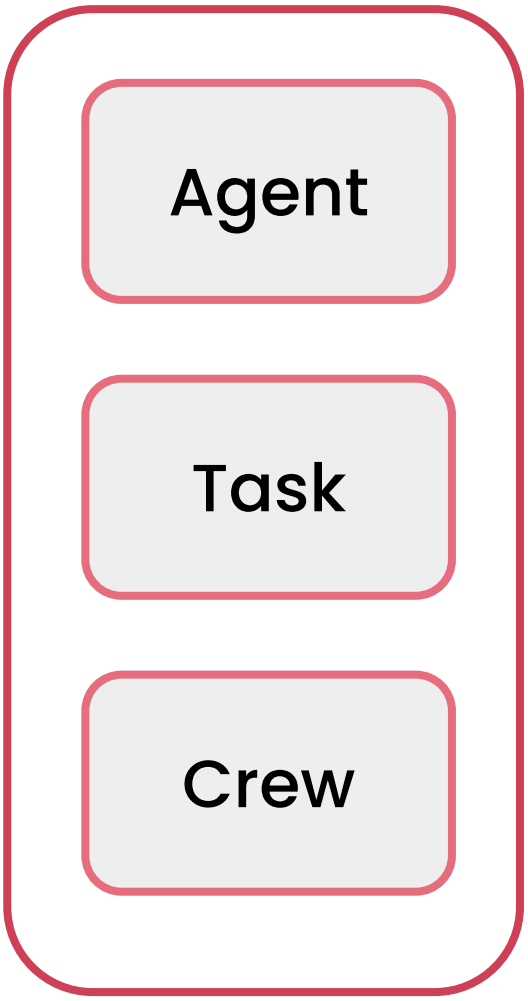

crewAI
crewAI
v0.22.4 9.8k 1.2k
crewAI
Home
Core Concepts
Agents
What is an Agent?
Agent Attributes
Creating an Agent
Conclusion
Tasks
Tools
Processes
Crews
Collaboration
Agents
What is an Agent?
What is an Agent?
An agent is an autonomous unit programmed to:
Perform tasks
Make decisions
Communicate with other agents
Think of an agent as a member of a team, with specific skills and a particular job to do. Agents can have different roles like 'Researcher', 'Writer', or 'Customer Support', each contributing to the overall goal of the crew.

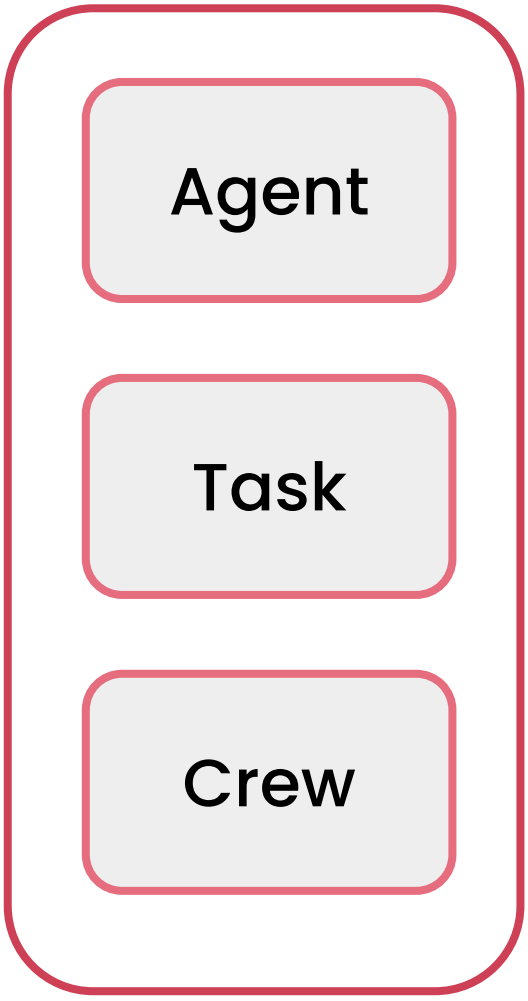

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v0.22.4 9.8k 1.2k
crewAI
Home
Core Concepts
Agents
Tasks
Tools
Processes
Crews
Collaboration
Collaboration Fundamentals
Enhanced Attributes for Improved Collaboration
Delegation: Dividing to Conquer
Implementing Collaboration and Delegation
Example Scenario
Conclusion
How to Guides
Tools Docs
Examples
Telemetry
Collaboration
Collaboration Fundamentals
Core of Agent Interaction
Collaboration in CrewAI is fundamental, enabling agents to combine their skills, share information, and assist each other in task execution, embodying a truly cooperative ecosystem.
Information Sharing: Ensures all agents are well-informed and can contribute effectively by sharing data and findings.
Task Assistance: Allows agents to seek help from peers with the required expertise for specific tasks.
Resource Allocation: Optimizes task execution through the efficient distribution and sharing of resources among agents.
Enhanced Attributes for Improved Collaboration
The Crew class has been enriched with several attributes to support advanced functionalities:
Language Model Management (manager_llm , function_calling_llm): Manages language models for executing tasks and tools, facilitating sophisticated agent-tool interactions. It's important to note that manager_llm is mandatory when using a hierarchical process for ensuring proper execution flow.
Process Flow (process): Defines the execution logic (e.g., sequential, hierarchical) to streamline task distribution and execution.
Verbose Logging (verbose): Offers detailed logging capabilities for monitoring and debugging purposes. It supports both integer and boolean types to indicate the verbosity level.
determines the kind of tasks the agent is best suited for.

Let’s recap!


Task
LLMs
Tools
Answer


Task
Task
LLMs
LLMs
Researcher
Writer


Agent
Task
Crew



- Agents perform better when role-playing
- You should focus on goals and expectations
- One Agent can do multiple tasks
- Tasks and Agents should be granular
- Tasks can be executed in different ways
- It's pretty easy to create multi-agents with crewAI

Lesson 3 starts here?

See wide slides for L3A
Lesson 3B is narrow slides for code, found here.

Multi AI Agent Systems with crewAI

Multi agent customer support automation (code)

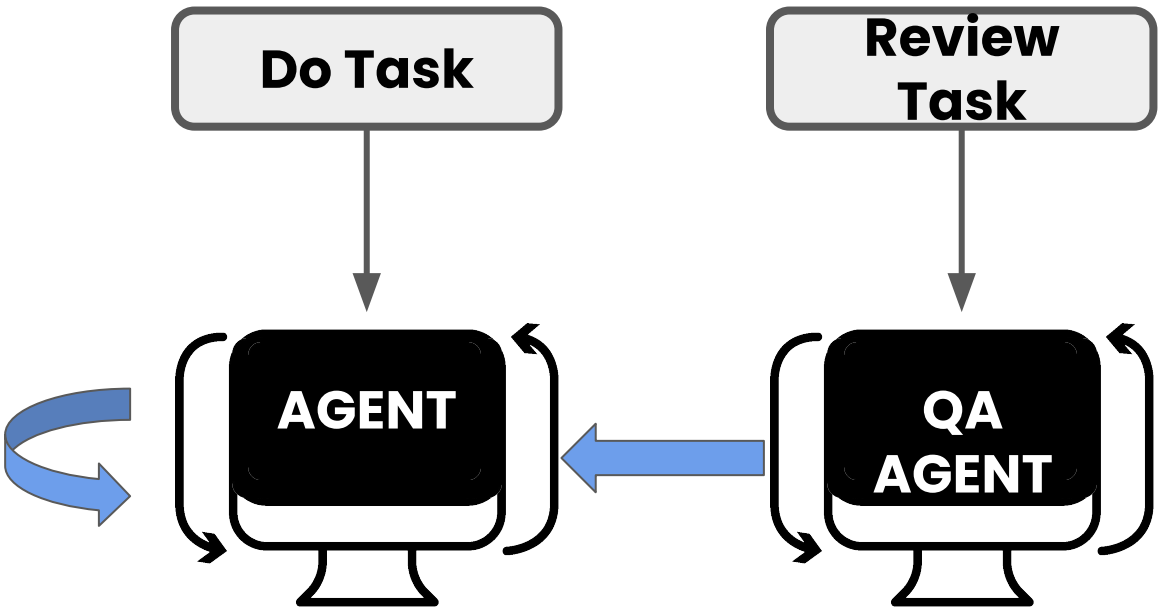




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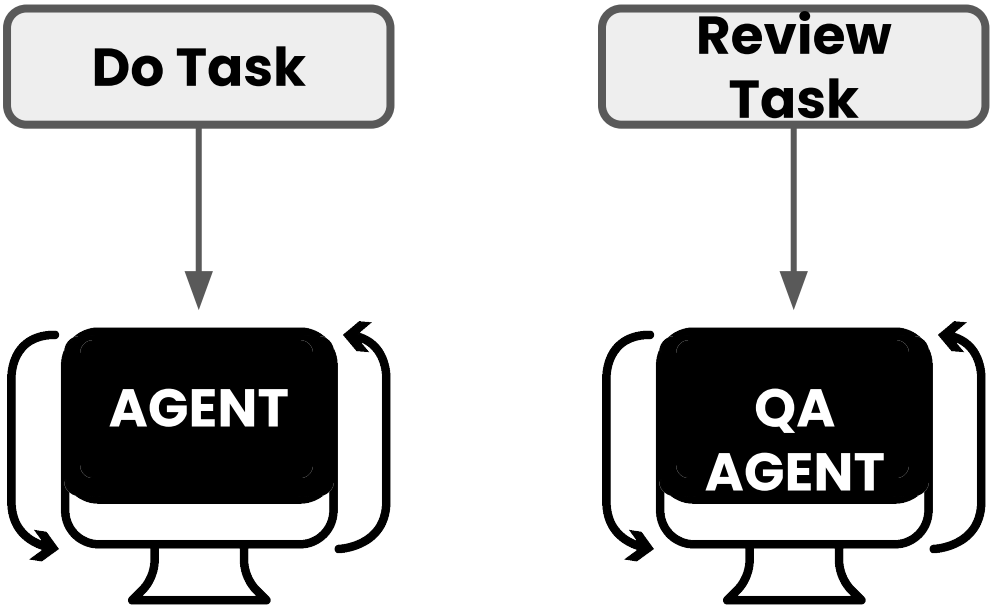
Allow Delegation



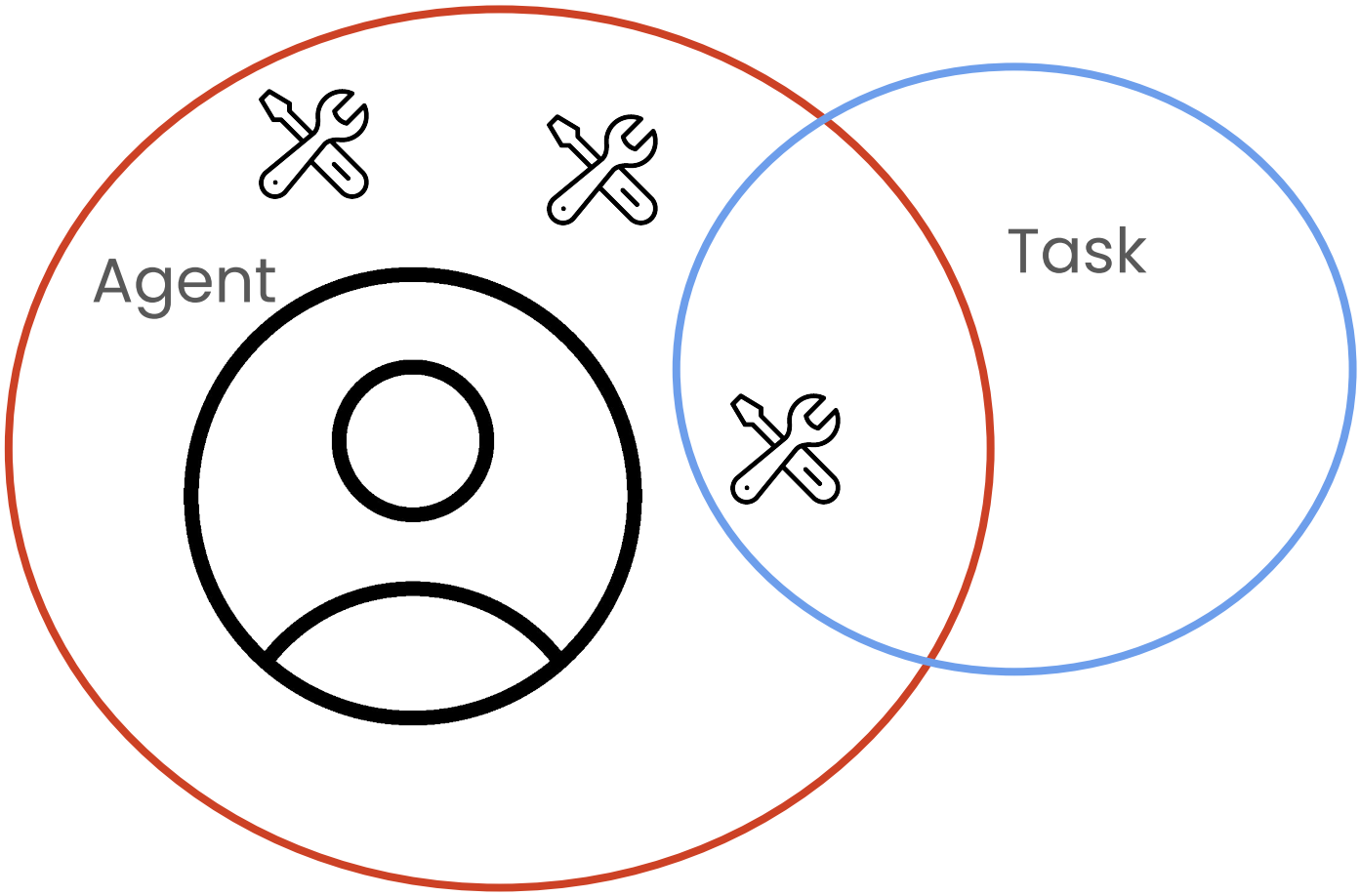
Not allowed to delegate to other agents (does its own work)

Can do its own work and delegate back to other agent

Design Pattern: Quality Assurance



Assigning Tools



Lesson 3C

see wide slides

Lesson 4A
(move to wide slides)

Lesson 4B

(code, so narrow slides)

Multi AI Agent Systems with crewAI

Tools for a customer outreach campaign (code)





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Exception Handling

- Option 1: Framework stops execution (you can choose how to handle the exceptions, such as restart execution.)
- Option 2: Framework continues execution (CrewAI is designed this way)

L4C (use wide slides)

[Jupyter Notebook]

Lesson 5B

(slides to be used within code walkthrough)

Multi AI Agent Systems with crewAI

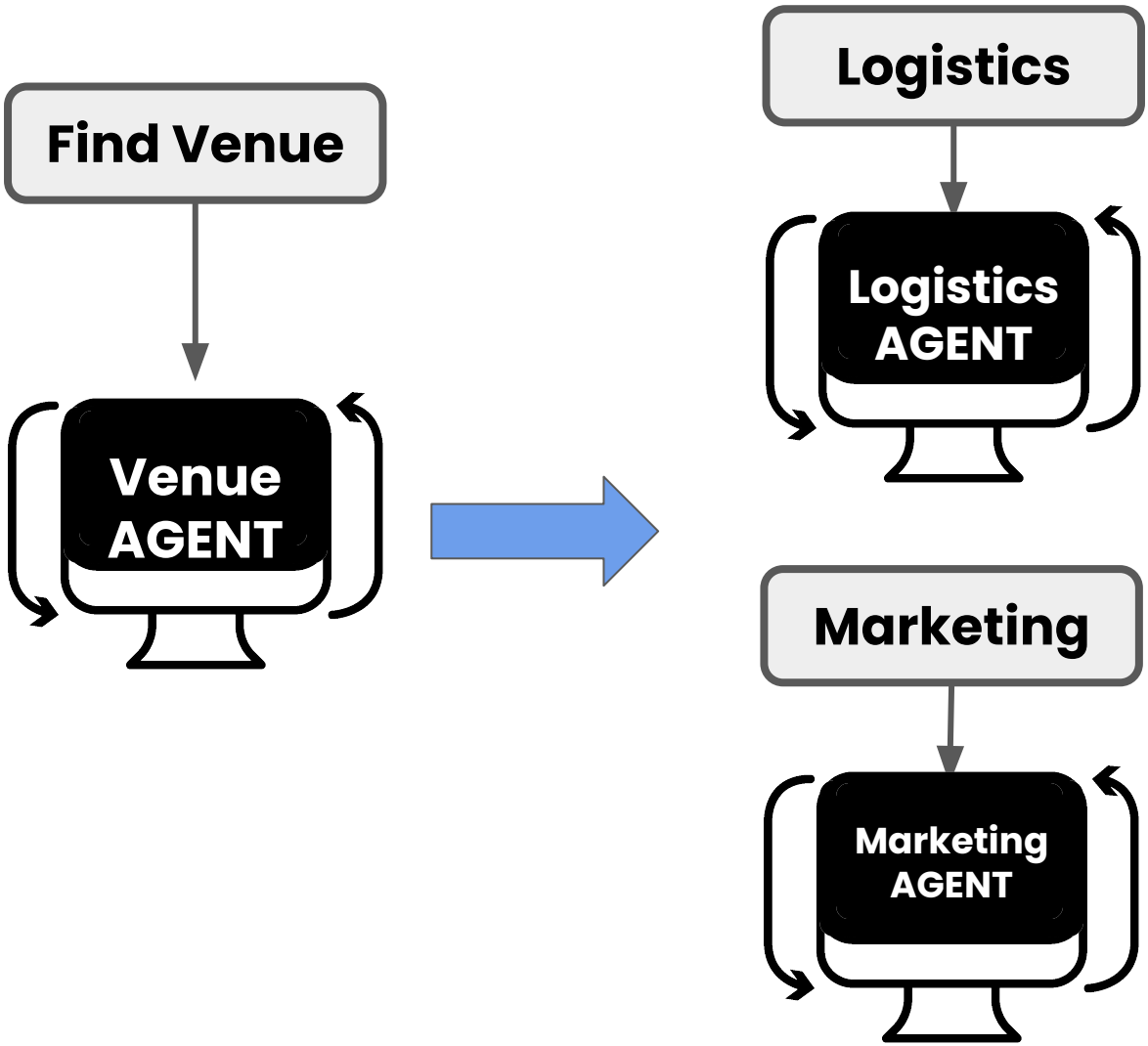
Automate event planning (code)





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Asynchronous Execution



Lesson 5C

(see wide slides)

Lesson 6A (see wide slides)
Lesson 6B (narrow slides here)

Multi AI Agent Systems with crewAI

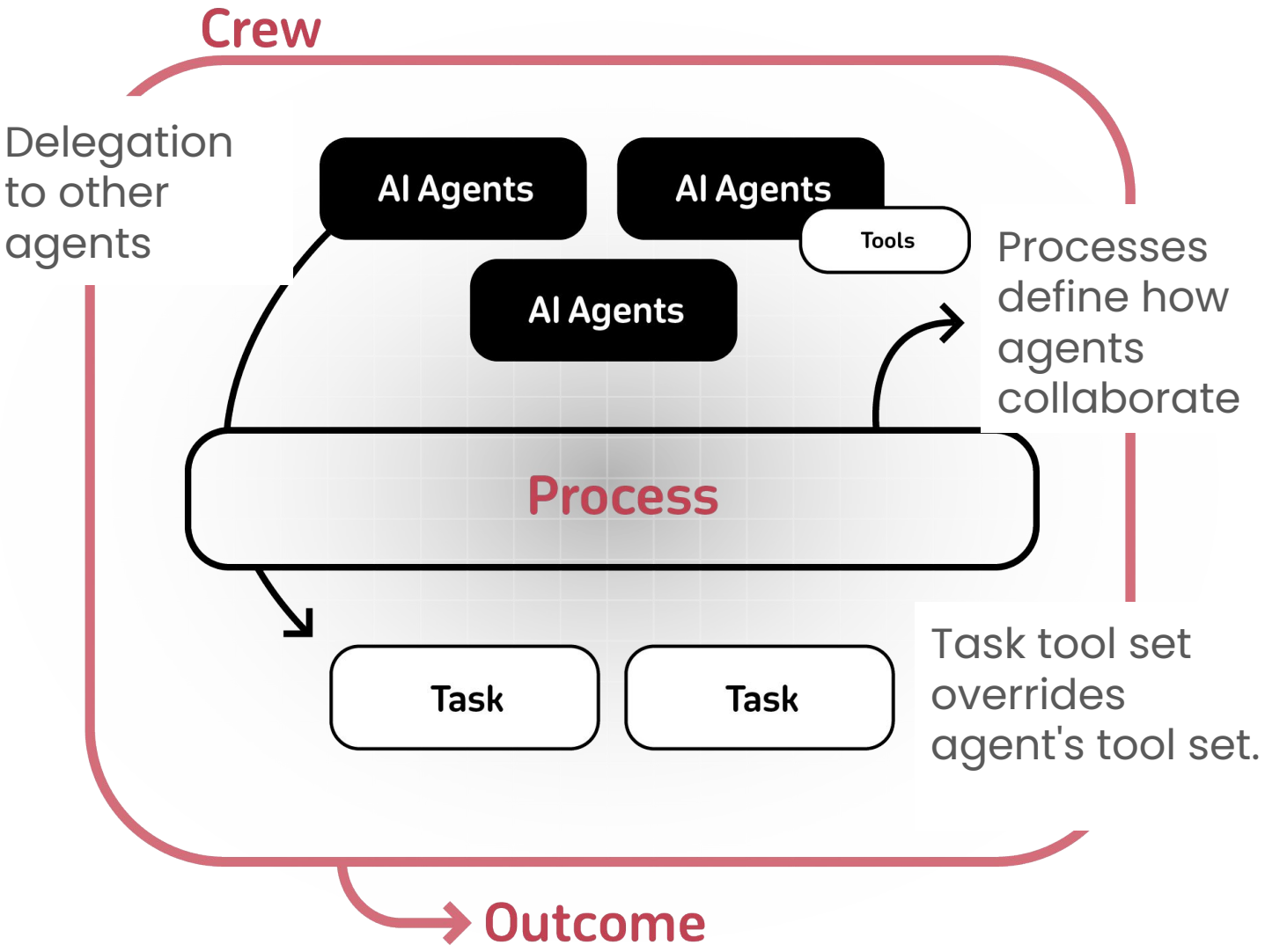
Multi agent collaboration for financial analysis (code)





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[Jupyter Notebook]


Crew
Delegation to other agents
AI Agents
AI Agents
Tools
AI Agents
Processes define how agents collaborate
Process
Task
Task
Task tool set overrides agent's tool set.
Outcome

Lesson 7

All in narrow slides

Multi AI Agent Systems with crewAI

Build a crew to tailor job applications





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Job Application Crew

your first multi-agent system

What is the Goal?

Maximize your chances of getting an interview for a given job posting

What is the Process?

- Learn about the job requirements
- Cross that with your skill set and experiences
- Reshape resume to highlight relevant areas
- Rewrite resume with appropriate language
- Set talking points for your initial interview